WASP-43b

R-0497

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_230326` · created 2026-07-20T14:04:20+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460029.8179 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.104 +/- 0.03
Transit depth (Rp/Rs)^2	1.08 +/- 0.62 [%]
Orbital Inclination (inc)	84.87 +/- 2.66
Airmass coefficient 1 (a1)	1.289 +/- 0.013
Airmass coefficient 2 (a2)	-0.1353 +/- 0.0044
Scatter in the residuals of the lightcurve fit is	1.01 %
Best Comparison Star	None
Optimal Aperture	4.24
Optimal Annulus	6.48
Transit Duration (day)	0.0532 +/- 0.0058

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.104 ± 0.03 vs 0.1594 (deviation 1.85σ)
Duration fit vs published	0.0532 d vs 0.0512 d (deviation 0.34σ)
Mid-time O–C	-4.0 min
Depth SNR	3.5
Residual scatter	1.01 %

Light curve

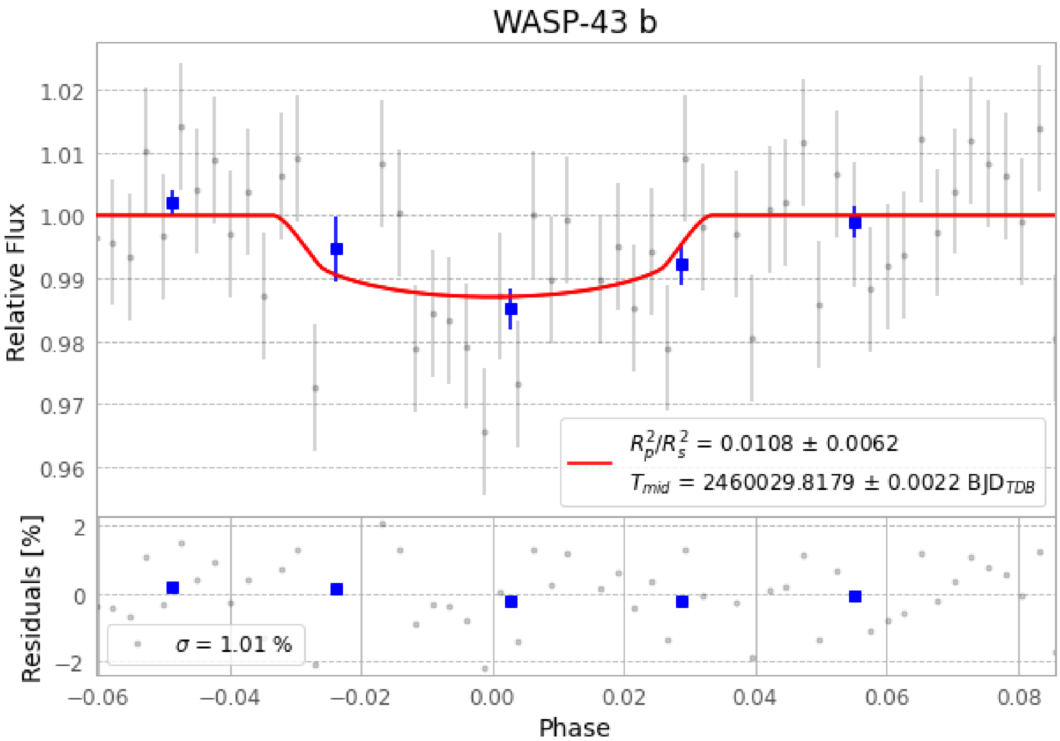

Figure 1 — FinalLightCurve_WASP-43 b_2023-03-25.png (SHA-256 b388a5f7e3fecee7…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [480, 233], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 9a5e2eb4cf066a5e…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

56 FITS frames (37 MB), `WASP-43230326061815.FITS` → `WASP-43230326090916.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230326.log (88dff7f2e313…), FinalLightCurve_WASP-43 b_2023-03-25.png (b388a5f7e3fe…), FinalLightCurve_WASP-43 b_2023-03-25.csv (b494afb65811…)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)